TITOgraphy: When Order Takes Shape

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ABSTRACT

Definition (Partial Order)**.** *A partially ordered set is set P together with a relation $\preceq$ on P that satisfies the following axioms:*

1. $a \preceq a, \forall a \in P$ (reflexivity).

2. For all $a, b \in P$, if $a \preceq b$ and $b \preceq a$, then $a = b$ (antisymmetry).

3. For all $a, b, c \in P$, if $a \preceq b$ and $b \preceq c$, then $a \preceq c$ (transitivity).

Definition (Total Order)**.** *Partial order $\preceq$ is a **total order** if for all $a, b \in P$, $a \preceq b$ or $b \preceq a$*

Definition (TITO)**.** *Fix $n \in \mathbb{Z}_{>0}$. A translation-invariant total order (TITO) is a total order $\preceq$ on $\mathbb{Z}$ such that for all $a, b \in \mathbb{Z}$, $a \preceq b$ if and only if $a + n \preceq b + n$.*

OBJECTIVE:

- Developing an algorithm to compare TITOs provides a concrete and computational way to study and visualize this structure.
- Implementing a comparator that outputs TITOs' relations and generates visualization.

The main goal of our project is to design and implement a Python package to compute TITO normalization, comparison, join and local Hasse diagrams.

THEORETICAL BACKGROUND

Definition (Block)**.** *A block of ($\preceq$) is a subset I of $\mathbb{Z}$ such that the ordering of I by ($\preceq$) has no minimal or maximal element, and for any $a, c \in I$, the interval $\{b \mid a \preceq b \preceq c\} \subset I$ is finite.*

Definition (Waxing/Waning)**.** *Block I is **waxing** if $x \preceq x + n$ for all $x \in I$. I is **waning** if $x + n \preceq x$ for all $x \in I$.*

Definition (Join)**.** *Least upper bound of 2 elements in the poset WO($TTot_n$).*

Definition (Inversion)**.** *An* inversion *of a TITO $(\prec)$ is a reflection index (a, b) with $b \prec a$. Denote the set of inversions of $(\prec)$ by $N(\prec)$.*

Definition (Weak Order)**.** $(\prec_1) \leq (\prec_2)$ *if and only if* $N(\prec_1) \subseteq N(\prec_2)$.

Example: (n=3)
$\ldots 4 \preceq 1 \preceq -2 \preceq -5 \ldots - 3 \preceq 2 \preceq 0 \preceq 5 \preceq 3 \preceq 8 \ldots$
Blocks:
1. $[\ldots 4 \preceq 1 \preceq -2 \preceq -5 \ldots] \implies$ window: $[\underline{1}]$
2. $[\ldots 2 \preceq 0 \preceq 5 \preceq 3 \ldots] \implies$ window: $[2, 0]$

Window notation: $[\underline{1}][2, 0]$

COMPARISON ALGORITHM

Main idea: decompose the inversion set of a TITO into disjoint subsets, and find comparable finite representations for each component.

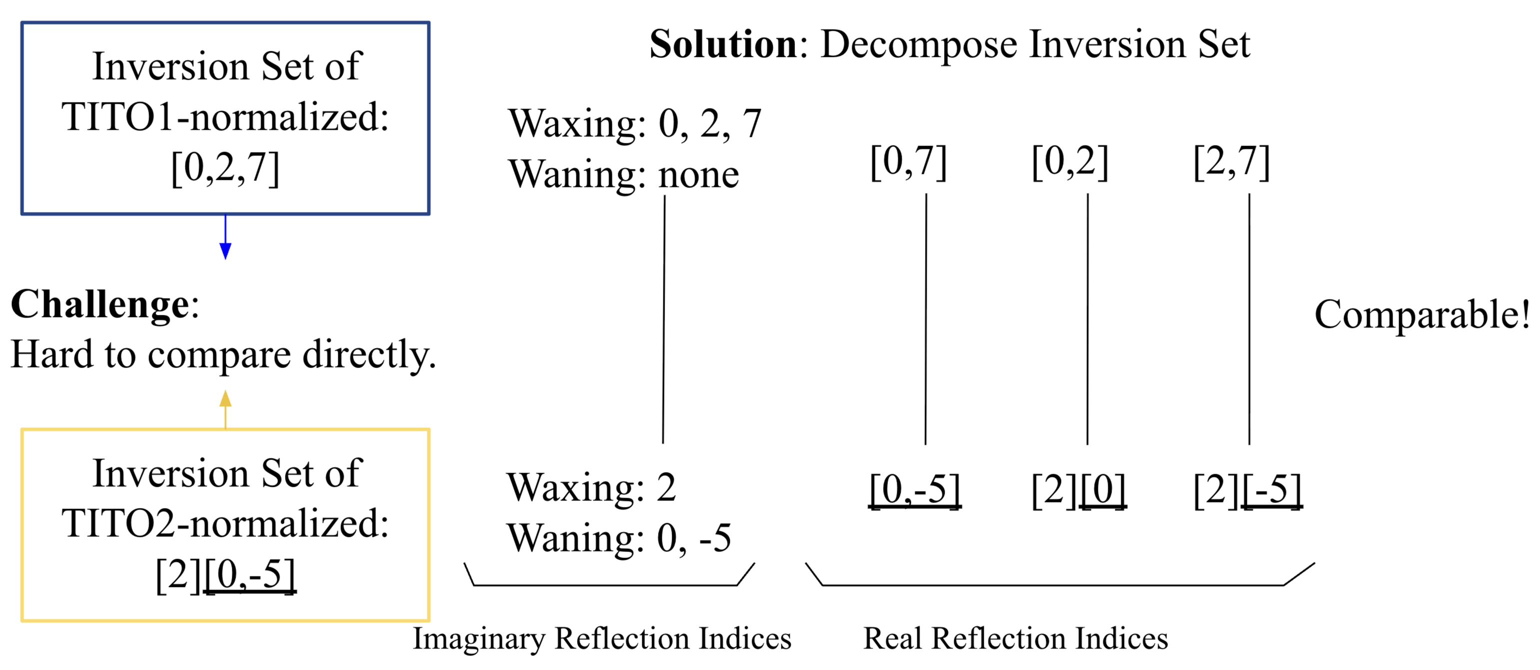

Figure 1: *Main idea behind the comparison algorithm. Here $n = 3$.*

NORMALIZATION ALGORITHM: One important pre-processing step in our algorithm is to normalize the window notation so that the following conditions are satisfied:

1. Within each window, the smallest residue class is always at the start.
2. Within each window, the first number is always between 0 and $n - 1$. This provides a unique normalized window notation for each TITO.

Imaginary Reflection Indices: Imaginary reflection indices come from determining, for each residue class, whether it lies in a waxing block or a waning block.
Real Reflection Indices: Real reflection indices come from comparing each pair of residue classes, which can be summarized in the following 6 cases:

Type	Block Normalized Format	Example	Inversion Representation	Inversion Size
Case 1	$[i][j]$, $i < j$	$[0][1]$	$\{(-2, 0), (-5, 0), \cdots\}$	Infinite
Case 2	$[i][j]$, $i > j$	$[1][0]$	$\{(0, 1), (0, 4), \cdots\}$	Infinite
Case 3	$[i, j]$, $i < j$	$[0, 7]$	$\{(0, 1), (0, 4)\}$	Finite
Case 4	$[i, j]$, $i > j$	$[0, -2]$	$\{(-2, 0)\}$	Finite
Case 5	$\underline{[i, j]}$, $i < j$	$\underline{[0, 4]}$	complement of $\{(0, 1), (0, 4)\}$	Cofinite
Case 6	$\underline{[i, j]}$, $i > j$	$\underline{[0, -5]}$	complement of $\{(-2, 0)\}$	Cofinite

Table 1: *Description of inversions generated by two residue classes in each case.*

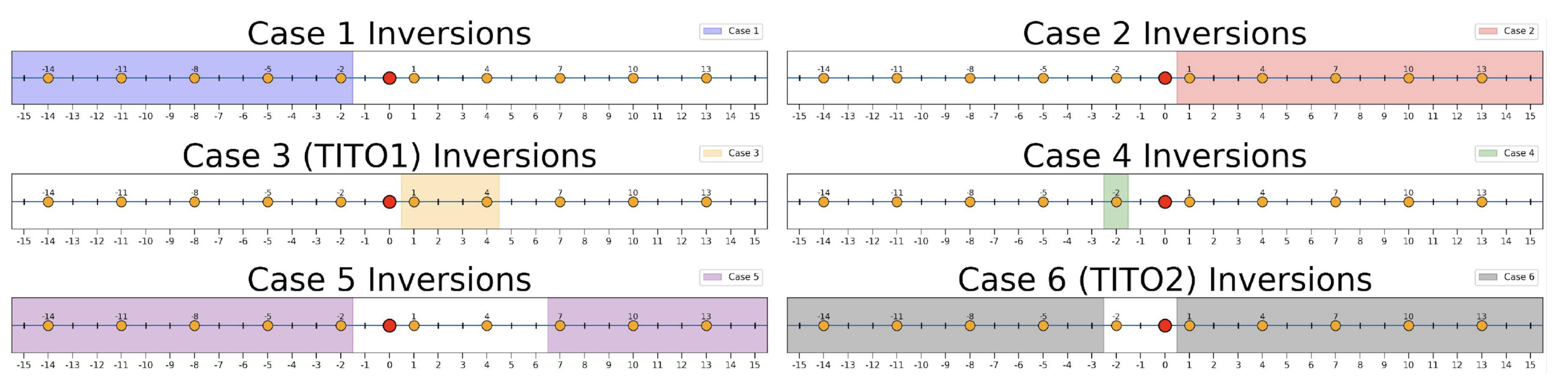

Figure 2: *Inversions generated by two residue classes.*

DIAGRAMS

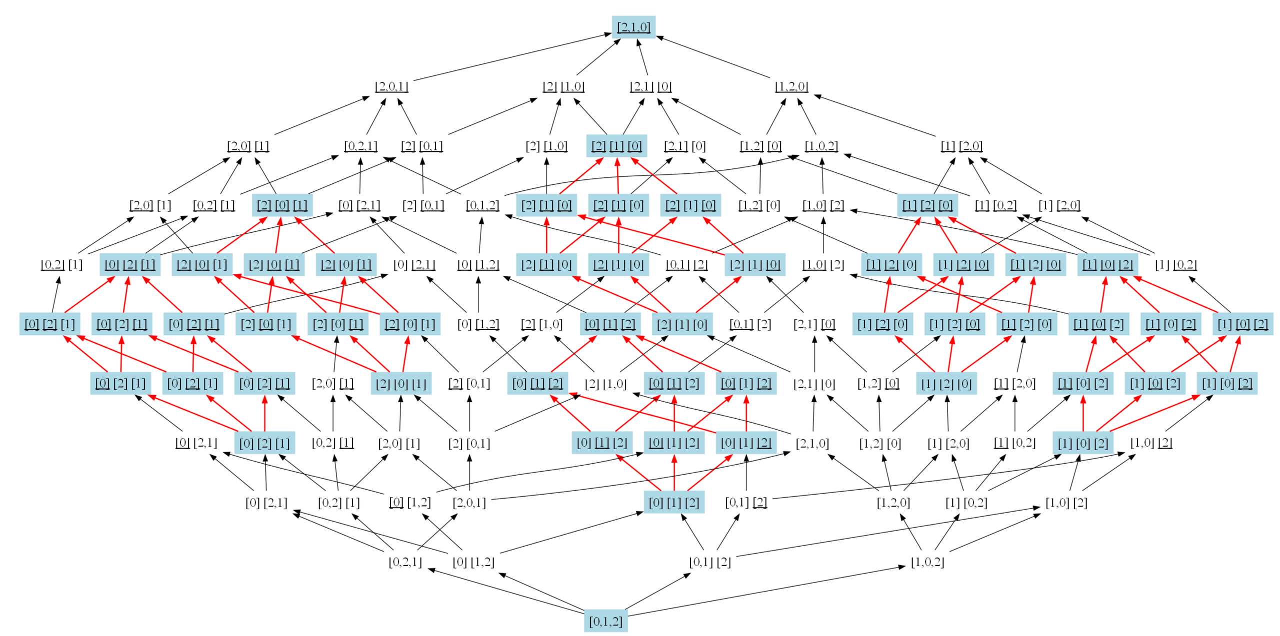

Figure 4: *Hasse Diagram for $n = 3$ with elements $0, 1, 2$.*

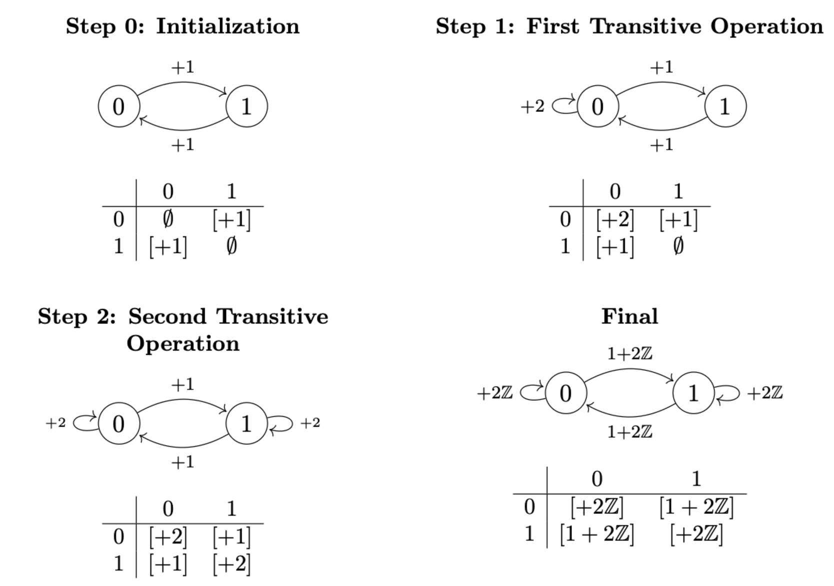

Figure 5: *Demonstration of finding the join of TITOs $[1, 0]$ and $[2, 1]$ with $n = 2$.*

CONCLUSION

We can use a finite representation for the infinite structure of the inversion set of TITO. We have accomplished a Python package:

- Normalizing a given TITO;
- Comparing given TITOs in weak order;
- Computing the join of two TITOs;
- Generating Hasse diagrams for small order-ideals of TITOs.

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